

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/02

Paper 2 Structured Questions

2 Sept 2015

Candidates answer on the Question Paper

2 hours

Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
A Data Booklet is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and
clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [] at the end of each
question or part question.

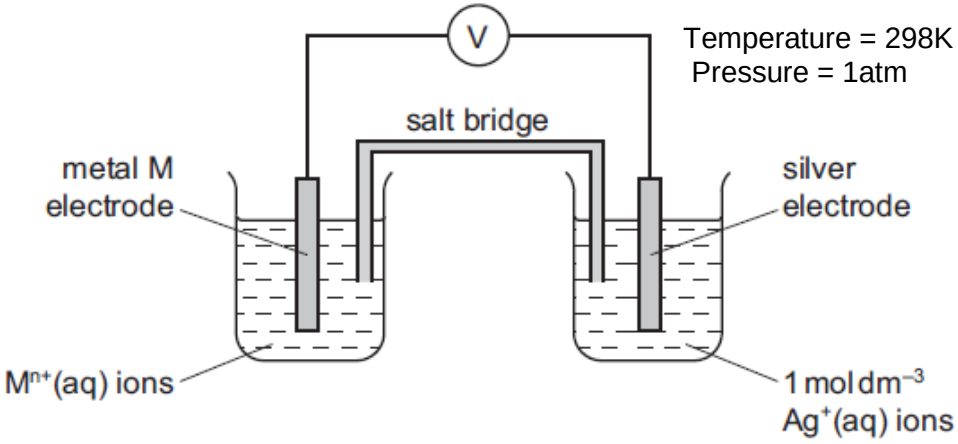
At the end of the examination, fasten all your work
securely together.

For Examiner's Use	
Section A	
1	10
2	17
3	12
4	16
5	15
Significant figures	
Handwriting	
Total	70

This document consists of **17** printed pages and **1** blank page.



Answer **ALL** questions on the spaces provided.

1	An experiment can be set up to identify an unknown metal M and the charge of its ions, M^{n+}. This is achieved by investigating how the cell potential containing this metal M in contact with an aqueous solution of its ions, $M^{n+}(aq)$ (where $n = 1, 2$ or 3), changes as $M^{n+}(aq)$ is diluted. It is known that the cell potential of a cell, E_{cell}, is related to the standard electrode potential, $E_{cell}^{\ominus}$, by the following equation. $E_{cell} = E_{cell}^{\ominus} - \frac{0.06 \log[M^{n+}(aq)]}{n}$ Since a standard hydrogen half-cell was not available, a standard half-cell consisting of silver in contact with a 1.0 mol dm^{-3} solution of silver nitrate was used to connect to the half-cell with metal M, in contact with $M^{n+}(aq)$.
(a)	Draw a labelled diagram you would set up in order to measure the cell potential of this electrochemical cell.
	
	• <u>salt bridge and voltmeter in closed circuit</u> • <u>correct half-cell diagrams ($1 \text{ mol dm}^{-3} \text{ Ag}^{+}(aq)$ must be indicated)</u>
	[2]
(b)	With relevant data from the <i>Data Booklet</i>, suggest whether the M^{n+}/M half-cell is the oxidation or reduction half-cell in (a). Explain your answer.
	Type of half-cell: <u>oxidation half-cell</u>
	Explanation:
	The <u>$E^{\ominus}(\text{Ag}^{+}/\text{Ag})$ being $+0.80\text{V}$ is the most positive amongst the metal/ion half cells</u>, the Ag^{+}/Ag half-cell is reduction half-cell. Hence, the M^{n+}/M half-cell is the oxidation half-cell.
	[1]

- (c) Write a plan for conducting a series of experiments to investigate how the cell potential changes as the aqueous solution of its ions, $M^{n+}(aq)$, is diluted.

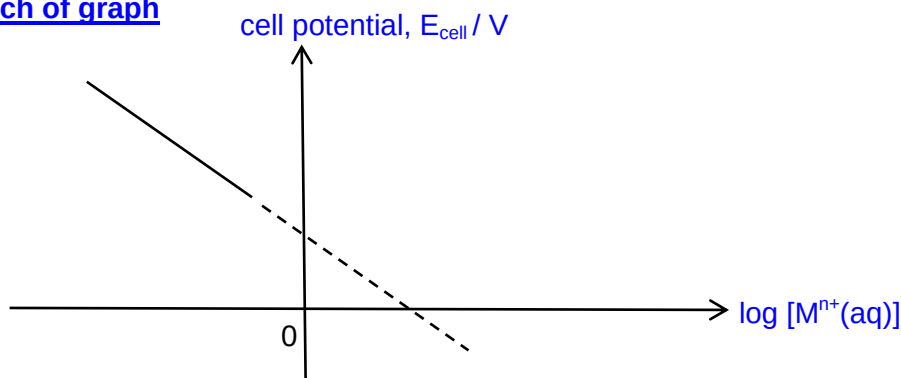
You may assume that you are provided with the following.

- 1.0 mol dm⁻³ solution of silver nitrate, AgNO₃(aq)
- 0.5 mol dm⁻³ solution of metal M ions, $M^{n+}(aq)$
- deionised water
- the apparatus normally found in a school or college laboratory

Your plan should include the following.

- details for the preparation of a suitable range of diluted solutions of $M^{n+}(aq)$ of accurate concentrations;
- description of how the electrochemical cell set-up in (a) is assembled;
- an outline of how results would be obtained;
- on the grid provided, a sketch of the graph that shows the relationship between $\log [M^{n+}(aq)]$ and the cell potential measured, E_{cell} ;
- an explanation of the shape of this graph;
- how the data obtained from this graph would be used to determine the
 - charge of metal ion, M^{n+}
 - the standard electrode potential for metal, M, $E^\ominus(M^{n+}/M)$.

1. Sketch of graph



Marking points for graph

Linear shape and decreasing trend of graph **with positive y-intercept (positive $E_{cell}^\ominus$)**

Explanation of shape of graph

As concentration of $M^{n+}(aq)$ increases, equilibrium position of $M^{n+}(aq) + ne^- \rightleftharpoons M(s)$ shifts to the right causing $E(M^{n+}/M)$ to be more positive.

Since $E_{cell} = E^\ominus(Ag^+/Ag) - E(M^{n+}/M) = +0.80 - E(M^{n+}/M)$, E_{cell} becomes less positive with increasing $\log [M^{n+}(aq)]$.

Manipulation of data obtained from graph

$$\text{Gradient} = \frac{-0.06}{n} \Rightarrow n = \frac{-0.06}{\text{gradient}}$$

$$\text{y-intercept} = E_{cell}^\ominus$$

$$\text{y-intercept} = E_{cell}^\ominus = +0.80 - E^\ominus(M^{n+}/M)$$

$$E^\ominus(M^{n+}/M) = +0.80 - \underline{E_{cell}^\ominus} = \underline{+0.80 - \text{y-intercept}}$$

Preparation of the standard solutions of $M^{n+}(aq)$

- Pipette 10.0 cm^3 of 0.5 mol dm^{-3} solution of metal ions, $M^{n+}(aq)$ to a 50 cm^3 volumetric flask.
- Top up to the mark with distilled water, stopper the volumetric flask and shake this solution to obtain a homogeneous solution. Label this solution obtained as **FA 2**.
- Repeat steps (i) to(ii) using the volumes of **FA2** to **FA4** shown in the table below to prepare **FA 3** to **FA5**.

Name of soln	Volume of solution to be diluted / cm^3	$[M^{n+}(aq)]/\text{mol dm}^{-3}$
FA 2	10.0 cm^3 of $0.5\text{ mol dm}^{-3} M^{n+}(aq)$	0.100
FA 3	10.0 cm^3 of FA2	0.0200
FA 4	10.0 cm^3 of FA3	0.00400
FA 5	10.0 cm^3 of FA4	0.000800

1(b) Procedure to record cell potential of electrochemical cell in (a)

- Using a 50 cm^3 measuring cylinder, transfer 40 cm^3 of 1.0 mol dm^{-3} solution of $\text{AgNO}_3(aq)$ to a 100 cm^3 beaker. Immerse the silver strip into the AgNO_3 solution.
- Using another 50 cm^3 measuring cylinder, transfer 40 cm^3 of 0.5 mol dm^{-3} solution of $M^{n+}(aq)$ to another 100 cm^3 beaker. Immerse the metal M strip into the M^{n+} solution.
- Set up the apparatus as shown in (a). Place the two half-cells in a thermostatically controlled water bath to maintain constant temperature of 298 K . Connect the electrodes to the voltmeter using crocodile clips. The salt bridge is prepared by soaking a filter paper strip in saturated KNO_3 solution.
- Recording the voltmeter reading immediately after the circuit is closed.
- Repeat steps (i) to (iv) using solutions **FA2** to **FA5**. Rinse the metal M and silver strips with deionised water and wipe it clean and dry before immersing into their respective solutions. The measuring cylinders are to be rinsed with the standard solutions of $M^{n+}(aq)$ before use.

Points to be included when writing procedures:

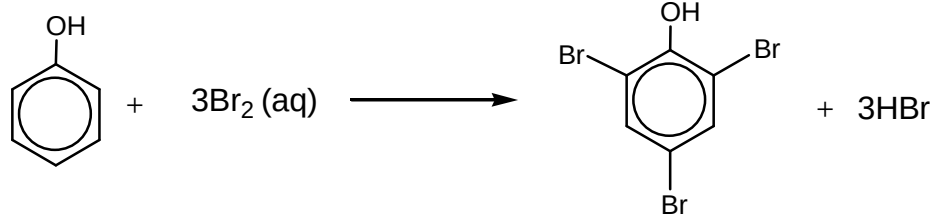
- Appropriate capacity and choice of apparatus (based on its level of precision)** eg burette with volumetric flask to be used for dilution
- Clear presentation of the concentrations of the diluted solutions with correct corresponding volumes of stock solutions** required for dilution eg. use of recording tables showing concentrations of the standard solutions prepared. Total number of diluted solutions should be at least 4.
- Clear and coherent procedures for set-up of electrochemical cell** citing how measurements of appropriate quantities of chemicals required for the half-cells are obtained.
- Precaution to ensure that the reduction half-cell is at standard condition upon repeating experiment** such as cleaning the metal electrodes and changing of electrolyte solutions including 1.0 mol dm^{-3} solution of silver nitrate before connecting the circuit.
- Recording of voltmeter reading immediately** at the instant **when the circuit is closed** to prevent fluctuations in the reading due to equilibrium shift as concentration of ions in electrolyte changes.

[9]

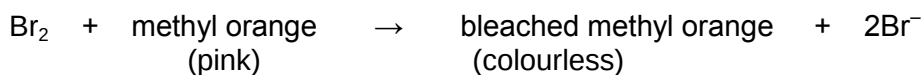
[Total: 12]

2	Nitrogen forms two stable anions, NO_3^- and NO_2^- . Bromine forms four stable anions, BrO^- , BrO_2^- , BrO_3^- and BrO_4^- .	
(a)	All group II nitrates decompose readily on heating, leaving a white residue.	
(i)	Write an equation showing the reaction that occurs when $\text{Ca}(\text{NO}_3)_2$ is heated. [1]	
	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	
(ii)	Explain the variation down the group of the ease of thermal decomposition of Group II metal nitrates. [2]	
	- the cation size <u>increases</u> causing the <u>charge density of the cation to decrease</u> - hence, the <u>polarising power</u> of cation <u>decreases</u> - extent of polarisation of electron cloud of NO_3^- anion <u>decreases</u>/ N-O bonds are weakened to a lower extent <u>more energy</u> is needed to break down the anion - thermal stability of nitrates <u>increases</u> - i.e. ease of thermal decomposition <u>decreases</u>.	
(b)	Bromate(V), BrO_3^- reacts with acidified hydrogen peroxide, H_2O_2 to form bromine, Br_2 .	
(i)	Write the half equation for the reduction of bromate(V), BrO_3^- ions. [1]	
	[R] $2\text{BrO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightarrow \text{Br}_2 + 6\text{H}_2\text{O}$	
(ii)	Hence, write the overall equation for the reaction between bromate(V) ions and acidified hydrogen peroxide. [1]	
	$2\text{BrO}_3^- + 5\text{H}_2\text{O}_2 + 2\text{H}^+ \rightarrow 5\text{O}_2 + \text{Br}_2 + 6\text{H}_2\text{O}$	
(iii)	The reaction can be catalysed by transition metal ions. What feature of transition metals such as iron makes them suitable as homogeneous catalysts in chemical reactions? [1]	
	Due to <u>similar energies of the 3d and 4s orbitals</u> , transition elements are able to show variable oxidation states.	
(c)	Bromine reacts with hydrogen in the following reaction. $\text{H}_2 + \text{Br}_2 \longrightarrow 2\text{HBr}$ This reaction is thought to occur by the following sequence of reactions. $\text{Br}_2 \rightleftharpoons 2\text{Br} \text{ (fast)}$ $\text{H}_2 + \text{Br} \longrightarrow \text{HBr} + \text{H} \text{ (slow)}$	

			$\text{H} + \text{Br}_2 \longrightarrow \text{HBr} + \text{Br} \text{ (fast)}$
		(i)	Explain what is meant by the term <i>order of a reaction</i> . [1]
			Order of reaction is the power which the concentration of the reactants is raised to in an experimentally determined rate equation. The overall order of the reaction is the sum of the individual orders. OR i.e. $\text{Rate} = k[\text{A}]^m[\text{B}]^n$ where m is the order of reaction with respect to [A] and n is the order of reaction with respect to [B]. m + n is the overall order of the reaction.
		(ii)	From the sequence of reactions given, show that the rate equation is: $\text{rate} = k [\text{H}_2] [\text{Br}_2]^{1/2}$ [2]
			From the slow step, the rate equation is: $\text{Rate} = k [\text{H}_2] [\text{Br}]$ As rate equation should be expressed in reactants, not intermediates, we need to replace [Br]. <u>OR</u> working/words to the same effect As equilibrium is reached for the first step: $K_c = \frac{[\text{Br}]^2}{[\text{Br}_2]}$ $[\text{Br}]^2 = K_c [\text{Br}_2]$ $[\text{Br}] = K_c^{1/2} [\text{Br}_2]^{1/2}$ Substituting back in to the rate equation: $\text{Rate} = k K_c^{1/2} [\text{H}_2] [\text{Br}_2]^{1/2} \text{ (shown)}$ Note: K_c can also be of the form $\frac{k_f}{k_b}$ where k_f is the rate constant of the forward reaction and k_b is the rate constant for the backward reaction in the rate equation.
		(d)	The kinetics of the reaction between bromate(V) and acidified bromide ions can be studied by the “clock” method, using phenol. The equation for the reaction is as follows. $\text{BrO}_3^-(\text{aq}) + 5\text{Br}^-(\text{aq}) + 6\text{H}^+(\text{aq}) \longrightarrow 3\text{Br}_2(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \quad \text{reaction I}$ When small but constant amounts of phenol and methyl orange is added to the reaction mixture, the bromine produced by the reaction will immediately react with the phenol present until all the phenol has been used up.
		(i)	Write a balanced equation for the reaction of phenol with aqueous bromine. [1]



As soon as the reaction I has produced sufficient bromine to react with the fixed amount of phenol, free bromine will appear in the solution. This bromine will immediately bleach the methyl orange solution causing the solution to turn from pink to colourless.



(ii) What is the simple relationship between the time taken for the solution to turn from pink to colourless and the initial rate of reaction I?

[1]

1/(time taken for the solution to turn from pink to colourless) is **directly proportional** to the **initial rate of reaction**.

A series of experiments was carried out using different volumes of the reagents, each solution being made up to the same total volume with water.

The following results were obtained.

Expt Number	Vol. of 0.50 moldm ⁻³ NaBrO ₃ /cm ³	Vol. of 0.10 moldm ⁻³ NaBr /cm ³	Vol. of 1.00 moldm ⁻³ H ₂ SO ₄ /cm ³	Vol. of phenol /cm ³	Vol. of methyl orange /cm ³	Vol. of water /cm ³	time taken for solution to turn from pink to colourless /s
1	15	10	5	3	2	75	140
2	30	10	5	3	2	60	70
3	15	15	5	3	2	70	94
4	15	10	15	3	2	65	16

(iii) Calculate the initial rate for each of these four experiments and use the results to deduce the order of the reaction with respect to each of the reactants. Explain your reasoning.

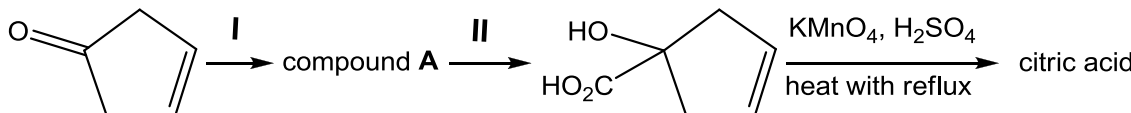
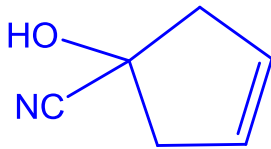
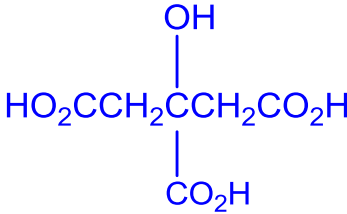
[4]

Initial rates $\propto$ 1/time taken

Expt Number	Initial rate/ s ⁻¹
1	0.00714
2	0.0142
3	0.0106
4	0.0625

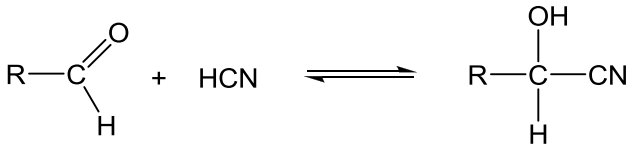
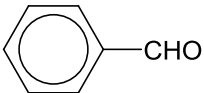
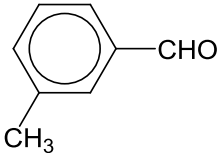
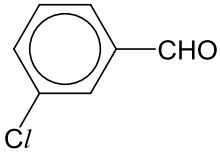
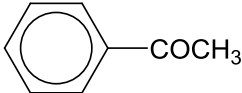
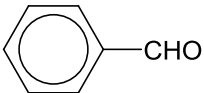
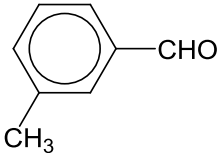
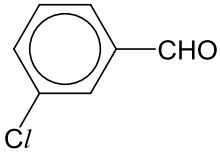
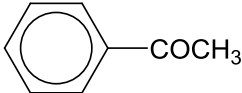
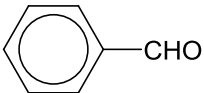
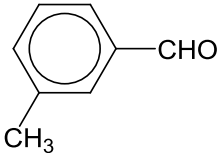
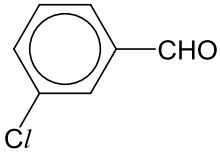
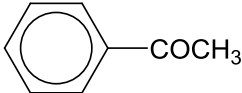
By inspection method:

		Comparing experiment 1 and 2, when $[\text{BrO}_3^-]$ is doubled and the concentration of the other reactants are constant, rate is doubled, this implies that the reaction is first order with respect to $[\text{BrO}_3^-]$. Comparing experiment 1 and 3, when $[\text{Br}^-]$ is increased by 1.5 times, and the concentration of the other reactants are constant, the rate also increases by 1.5 times, this implies that the reaction is first order with respect to $[\text{Br}^-]$. Compare experiment 3 and 4. Let the order of reaction with respect to $[\text{H}^+]$ be n $\frac{[15]^1[15]^1[5]^n}{[15]^1[10]^1[15]^n} = \frac{0.0106}{0.0625}$ $\left[\frac{1}{3}\right]^n \times \frac{3}{2} = \frac{1}{6}$ $\left[\frac{1}{3}\right]^n = \frac{1}{9} \quad n = 2$ Hence, the order of reaction with respect to $[\text{H}^+]$ is 2. The reaction is second order with respect to $[\text{H}^+]$.
		(iv) Hence write the overall rate equation for the reaction, stating the units for the rate constant. [2]
		Rate = $k [\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2$ Since Rate is always taken to be $\text{mol dm}^{-3} \text{s}^{-1}$ Units for $k = \text{mol}^{-3} \text{dm}^9 \text{s}^{-1}$ OR Since units for rate = s^{-1} in this experiment, Units for $k = \text{mol}^{-4} \text{dm}^{12} \text{s}^{-1}$
		[Total: 17]

3	Citric acid is a commodity chemical and more than a million ton is produced every year by fermentation. It is a natural preservative which is present in citrus fruits. Citric acid can be synthesised from cyclopent-3-enone by the following route: 																		
(a)	(i)	Suggest the reagents and conditions for steps I and II.																	
		Step I: HCN, trace amount of NaCN, 10°C to 20°C Step II: Dilute H₂SO₄, heat with reflux																	
	(ii)	Draw the structural formula of compound A and citric acid.																	
		 compound A	 citric acid																
[4]																			
A carboxylic acid present in unripe fruits is a colourless crystalline solid, V, which has the following composition by mass: C, 35.8%; H, 4.5%; O, 59.7%.																			
(b)	(i)	Given that the M_r of V is 134, use this value and the above information to determine the molecular formula of V .																	
		<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <th style="text-align: left;">Elements</th><th>C</th><th>H</th><th>O</th></tr> <tr> <td style="text-align: left;">Mass in 100g /g</td><td>35.8</td><td>4.5</td><td>59.7</td></tr> <tr> <td style="text-align: left;">No. of moles</td><td>$\frac{35.8}{12.0} = 2.983$</td><td>$\frac{4.5}{1.0} = 4.5$</td><td>$\frac{59.7}{16.0} = 3.731$</td></tr> <tr> <td style="text-align: left;">Simplest ratio</td><td>4</td><td>6</td><td>5</td></tr> </table> Empirical formula = <u>C₄H₆O₅</u> $n(Mr \text{ of } C_4H_6O_5) = 134$ $n [12.0(4) + 1.0(6) + 16(5)] = 134$ <u>n = 1</u> Hence, molecular formula of V is <u>C₄H₆O₅</u> Molecular formula of V <u>C₄H₆O₅</u>		Elements	C	H	O	Mass in 100g /g	35.8	4.5	59.7	No. of moles	$\frac{35.8}{12.0} = 2.983$	$\frac{4.5}{1.0} = 4.5$	$\frac{59.7}{16.0} = 3.731$	Simplest ratio	4	6	5
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[2]																			
A sample of V of mass 1.97 g was dissolved in water and the resulting solution titrated with 1.00 mol dm⁻³ NaOH. 29.4 cm³ were required for complete neutralization.																			
(c)	(i)	Use these data to deduce the number of carboxylic acid groups present in one molecule of V .																	

10

			Amount of NaOH used in reaction = $\frac{29.4}{1000} \times 1.00 = 0.0294 \text{ mol}$ Amount of V used in reaction = $\frac{1.97}{134} = 0.0147 \text{ mol}$ Ratio of V : NaOH = 0.0147 : 0.0294 = 1 : 2 Therefore, there are 2 carboxylic acid groups present in one molecule of V.
		(ii)	Compound V does not decolourise acidified KMnO_4 when heated. Compound W, an isomer of compound V, can be dehydrated to give a mixture of two isomeric alkenedioic acids, compound X and Y with molecular formula $\text{C}_4\text{H}_4\text{O}_4$. Gently heating the anhydrous crystals of compound Y produces a neutral compound Z, $\text{C}_4\text{H}_2\text{O}_3$, which neither reacts with sodium metal nor gives a precipitate with 2,4-dinitrophenylhydrazine. Suggest the identities of V, W, X, Y and Z.
			 compound V compound W compound X compound Y compound Z
			[6]
			[Total: 12]

4	(a)	This question is about a series of experiments conducted on carbon compounds and its derivatives in the laboratory. The uncatalysed addition reaction of HCN to a carbonyl group to give a cyanohydrin can be represented as a reaction in equilibrium as follows:  carbonyl compound cyanohydrin The K_c expression is given as $K_c = \frac{[\text{cyanohydrin}]}{[\text{carbonyl compound}][\text{HCN}]}$ The value of K_c is determined under identical conditions for the following carbonyl compounds. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>no.</th><th>carbonyl compound</th><th>$K_c / \text{mol}^{-1} \text{dm}^3$</th></tr> </thead> <tbody> <tr> <td>1</td><td></td><td>210</td></tr> <tr> <td>2</td><td></td><td>175</td></tr> <tr> <td>3</td><td></td><td>500</td></tr> <tr> <td>4</td><td>CH_3COCH_3</td><td>30</td></tr> <tr> <td>5</td><td></td><td>0.8</td></tr> </tbody> </table>	no.	carbonyl compound	$K_c / \text{mol}^{-1} \text{dm}^3$	1		210	2		175	3		500	4	CH_3COCH_3	30	5		0.8
no.	carbonyl compound	$K_c / \text{mol}^{-1} \text{dm}^3$																		
1		210																		
2		175																		
3		500																		
4	CH_3COCH_3	30																		
5		0.8																		
	(i)	Explain the differing K_c values for carbonyl compounds 1, 2 and 3. In carbonyl compound 2, the <u>electron-donating CH_3 group will make the carbonyl carbon less electron deficient and less susceptible to nucleophilic attack.</u> In carbonyl compound 3, the <u>electron-withdrawing Cl atom will make the carbonyl carbon more electron deficient and more susceptible to nucleophilic attack.</u> The <u>more electron-deficient the carbonyl carbon is</u>, (the more susceptible it will be to nucleophilic attack), <u>the more the equilibrium position lies to the right, the larger the K_c value.</u>																		

		(ii)	Explain why the value of K_c for carbonyl compound 5 is less than that for 4. <u>In carbonyl compound 5, there is the presence of a bulky phenyl group as compared to the CH_3 group in carbonyl compound 4. This causes steric hindrance to the attack of CN^- on the electron deficient carbonyl carbon in Step I.</u>																				
			[5]																				
	(b)	In another reaction, the thiocyanate ion, SCN^- is used to test for the presence of iron(III) ions. $\text{Fe}(\text{H}_2\text{O})_6^{3+}(\text{aq}) + \text{SCN}^-(\text{aq}) \rightleftharpoons [\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$																					
		(i)	Write the K_c expression for the reaction, stating its units. $K_c = \frac{[\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}}{[\text{Fe}(\text{H}_2\text{O})_6^{3+}][\text{SCN}^-]} \quad \text{mol}^{-1} \text{ dm}^3$																				
		(ii)	When equal volumes of $8.00 \times 10^{-3} \text{ mol dm}^{-3}$ of FeCl_3 and $8.00 \times 10^{-3} \text{ mol dm}^{-3}$ of KSCN are mixed and allowed to reach equilibrium, the resulting concentration of $[\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}(\text{aq})$ is $3.31 \times 10^{-3} \text{ mol dm}^{-3}$. Use this data to calculate the value of the equilibrium constant, K_c . <table border="1"><thead><tr><th></th><th>$\text{Fe}(\text{H}_2\text{O})_6^{3+}(\text{aq})$</th><th>$\text{SCN}^-(\text{aq})$</th><th>$\rightleftharpoons [\text{Fe}(\text{SCN})]^{2+}(\text{aq})$</th><th>$\text{H}_2\text{O}(\text{l})$</th></tr></thead><tbody><tr><td>Initial conc</td><td>4.00×10^{-3}</td><td>4.00×10^{-3}</td><td>0</td><td>-</td></tr><tr><td>Change</td><td>-3.31×10^{-3}</td><td>-3.31×10^{-3}</td><td>$+3.31 \times 10^{-3}$</td><td>-</td></tr><tr><td>Eqm conc</td><td>6.9×10^{-4}</td><td>6.9×10^{-4}</td><td>3.31×10^{-3}</td><td>-</td></tr></tbody></table> <u>[Note: ICE table is not a marking point, as long as students are able to get the eqm conc correctly by other methods]</u> $K_c = \frac{[\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}}{[\text{Fe}(\text{H}_2\text{O})_6^{3+}][\text{SCN}^-]} = \frac{3.31 \times 10^{-3}}{(6.9 \times 10^{-4})^2} = 6.95 \times 10^3 \text{ mol}^{-1} \text{ dm}^3$		$\text{Fe}(\text{H}_2\text{O})_6^{3+}(\text{aq})$	$\text{SCN}^-(\text{aq})$	$\rightleftharpoons [\text{Fe}(\text{SCN})]^{2+}(\text{aq})$	$\text{H}_2\text{O}(\text{l})$	Initial conc	4.00×10^{-3}	4.00×10^{-3}	0	-	Change	-3.31×10^{-3}	-3.31×10^{-3}	$+3.31 \times 10^{-3}$	-	Eqm conc	6.9×10^{-4}	6.9×10^{-4}	3.31×10^{-3}	-
	$\text{Fe}(\text{H}_2\text{O})_6^{3+}(\text{aq})$	$\text{SCN}^-(\text{aq})$	$\rightleftharpoons [\text{Fe}(\text{SCN})]^{2+}(\text{aq})$	$\text{H}_2\text{O}(\text{l})$																			
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Eqm conc	6.9×10^{-4}	6.9×10^{-4}	3.31×10^{-3}	-																			
		(iii)	The numerical value of K_c for this reaction was found to be 138 when the equilibrium mixture was placed in a hot water bath. With reference to your answer in (ii), deduce whether the reaction is exothermic or endothermic. <u>When temperature increases, the value of K_c decreases.</u> <u>This shows that the equilibrium position has shifted to the left to favour the endothermic reaction to absorb additional heat. Hence the forward direction is exothermic.</u>																				
			[6]																				

	(c) A student wished to determine the standard enthalpy change of reaction between methane and chlorine to form carbon tetrachloride, $\Delta H_1^\ominus$. $\text{CH}_4(\text{g}) + 4\text{Cl}_2(\text{g}) \longrightarrow \text{CCl}_4(\text{l}) + 4\text{HCl}(\text{g}) \quad \Delta H_1^\ominus$
	(i) Suggest the conditions required for this reaction to occur.
	uv light
	(ii) A teacher commented that despite having the appropriate conditions for the reaction to take place, it is still not possible to determine $\Delta H_1^\ominus$ directly in the laboratory. Suggest why this is so.
	A mixture of products will be formed./it is impossible to control the extent of the free radical substitution etc
	(iii) The teacher proceeded to provide the following information for the student to determine the enthalpy change of reaction, $\Delta H_1^\ominus$. $\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \longrightarrow \text{CH}_4(\text{g}) \quad \Delta H_2^\ominus$ $\frac{1}{2}\text{H}_2(\text{g}) + \frac{1}{2}\text{Cl}_2(\text{g}) \longrightarrow \text{HCl}(\text{g}) \quad \Delta H_3^\ominus$ By means of an energy cycle or otherwise, determine what other information is needed to calculate a value for $\Delta H_1^\ominus$.
	Any other suitable method such as algebraic method and correct energy cycle, <u>need to come with state symbols</u> and <u>balanced equations</u> Alternative answer: $\begin{aligned} \Delta H_1^\ominus &= \sum \Delta H_f^\ominus (\text{products}) - \sum \Delta H_f^\ominus (\text{reactants}) \\ &= [\Delta H_f^\ominus (\text{CCl}_4) + 4 \Delta H_3^\ominus] - \Delta H_2^\ominus \end{aligned}$ ----- $\Delta H_4^\ominus$ which is the <u>standard enthalpy change of formation of $\text{CCl}_4(\text{l})$</u> is required. ** students may opt to solve this questionn by algebraic method.
	[5]
	[Total: 16]

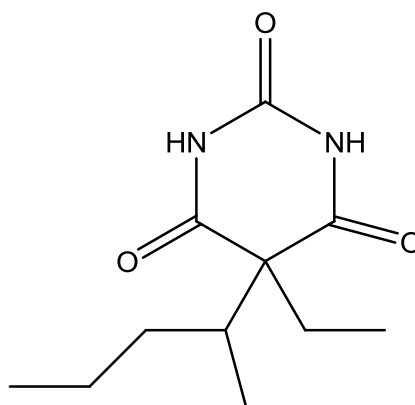
- 5 (a) (i) Describe the conditions needed to hydrolyse amides in the laboratory.

Acidic hydrolysis: HCl(aq) or $\text{H}_2\text{SO}_4\text{(aq)}$, heat with reflux

OR

Alkaline hydrolysis: NaOH(aq) , heat with reflux

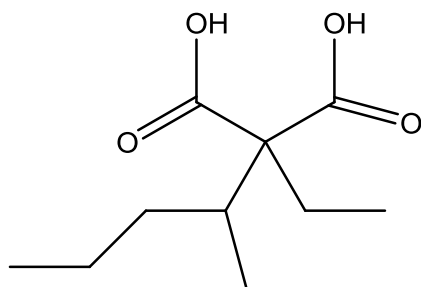
- (ii) Predict the products of the hydrolysis of the amide groups in the following drug.



Pentobarbital

[3]

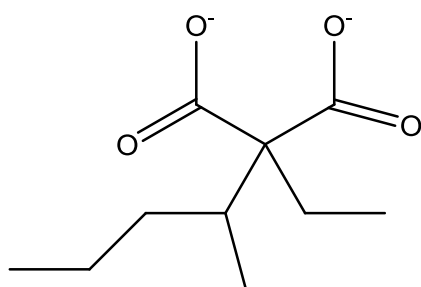
If student use acidic hydrolysis [answer here must be tagged to their answer in (i)]:



[1] NH_4^+ and CO_2

OR

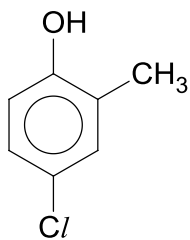
If student use alkaline hydrolysis [answer here must be tagged to their answer in (i)]:



NH_3 and HCO_3^- or CO_3^{2-}

- (b) Compound **S** can be formed from 2-methylphenol

15

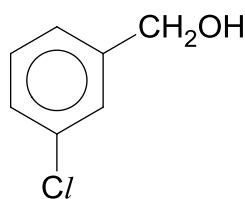
**S**

- (i) State the reagents and conditions required for the conversion.

.....

$\text{Cl}_2(\text{g})$ OR Cl_2 in CCl_4 , room temperature, in the dark

- (ii) Describe a simple chemical test that can be used to distinguish **S** from its isomer, **T**. State the reagents and conditions used and the observation for both compounds.

**T**

[3]

.....

.....

.....

$\text{K}_2\text{Cr}_2\text{O}_7$, dil H_2SO_4 , heat;

T: orange solution turns green, white ppt seen

S: orange solution remains;

or

$\text{Br}_2(\text{aq})$ room temperature;

S: Orange Br_2 decolourise, white ppt seen, white fumes

T: Orange Br_2 remains

or

$\text{PCl}_5(\text{s})$, room temp or $\text{SOCl}_2(\text{l})$, warm

T will give white fumes. S will not give white fumes. ;

or

neutral $\text{FeCl}_3(\text{aq})$, room temp

T will give purple/violet colouration. S will not give purple/violet colouration.

- (c) Compound **C** has the molecular formula $\text{C}_8\text{H}_6\text{O}_4$. It is an aromatic compound which contains three functional groups.

Data about the reactions of **C** are given in the table.

reaction	reagent	result
1	neutral $\text{FeCl}_3(\text{aq})$	violet complex formed
2	$\text{Br}_2(\text{aq})$ in an excess	Br_2 is decolourised; white solid formed with $M_r = 323.8$
3	2,4-dinitrophenylhydrazine	orange precipitate formed
4	PCl_5	white fumes and a neutral cyclic compound D formed with $M_r = 148$
5	Na	colourless gas evolved.
6	$[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$	No silver mirror seen

- (i) Name the functional group that reaction **1** shows to be present in **C**.

.....
Phenol

- (ii) Name the functional group that reaction **3** and **6** show to be present in **C**.

.....
Ketone

- (iii) 0.1 mol of **C** is reacted with excess sodium metal to produce hydrogen gas. The gas formed occupied a volume of 2.45 dm^3 at room temperature and pressure. Find the number of moles of hydrogen gas formed.

$pV=nRT$

$$1.01 \times 10^5 \times \frac{2.45}{1000} = n \times 8.31 \times 298$$

$n = 0.100 \text{ mol}$

- (iv) Based on reaction **5** and your answer in (i), (ii) and (iii), deduce, with reasoning, the identity of the third functional group present in **C**.

Amount of H_2 formed from 0.10 mol of **C** = 0.10 mol

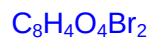
Mole ratio of **C** : H_2 = 1 : 1

1 mole of **C** reacts with Na to give 1 mole of H_2 , **C** contains 2 OH groups.

C contains phenol (1 OH), it must contain either 1 ROH or 1 CO_2H group that reacted with Na.

As **C** contains 4 O atoms, 1 O atom is from phenol, 1 O atom is from ketone, there are 2 O atoms left for the third functional group. Thus, the third functional group in **C** is carboxylic acid.

- (v) Deduce the molecular formula of the product formed in reaction 2.

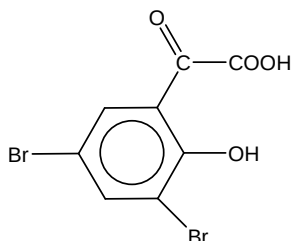


M_r of compound **C** = 166

M_r of product = 323.8

Difference in M_r = 157.8

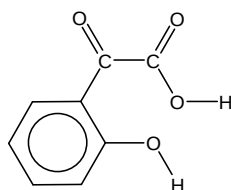
Number of Br atoms in product = 2



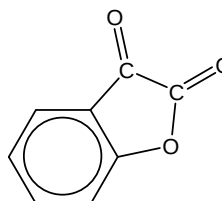
- (vi) Draw the fully displayed structures of **C** and **D**.

[9]

C:



D:



Note that displayed formula requires the showing of all bonds except the C & H on benzene rings.

[Total: 15]